

AKSHAT MITTAL

Generative AI & Python Backend Developer

akshatmittal2223@gmail.com | +91 9211082249 | New Delhi, India | [linkedin.com/in/akshat2223](https://www.linkedin.com/in/akshat2223) | github.com/AKSHAT-MITTAL22

SUMMARY

Python and Generative AI developer with hands-on experience building **LLM-integrated backend systems** using FastAPI, Flask, and Azure. Delivered a production automation pipeline that **cut manual processing effort by 80%** for a banking client. Currently deepening applied expertise in **RAG architecture, vector databases, and LangChain-based workflows** to move from API integration toward full GenAI system design - with a consistent track record of shipping working systems, not just prototypes.

CORE SKILLS

Languages & Frameworks: Python, SQL, FastAPI, Flask, REST API Design

Generative AI: LLM API Integration, Prompt Engineering, LangChain, Azure OpenAI, Agent-based Workflows

Cloud & Infrastructure: Microsoft Azure, Azure Blob Storage, Key Vault, Azure OCR

Engineering Practices: Application Architecture, Database Integration, Data Pipeline Automation, Agile/Cross-functional Delivery

WORK EXPERIENCE

Systems Engineer, Tata Consultancy Services (TCS)

Feb 2025 – Present | Noida, India

- Accomplished **20% faster delivery timelines** across cross-functional agile sprints by restructuring backend workflows with **Python and LLM-based automation (TCS AI Wisdom Next)**, aligning delivery with enterprise standards.
- Designed and deployed **REST APIs** enabling seamless data exchange across multiple business applications, reducing manual integration overhead between teams.
- Optimized scalable backend systems using Python frameworks and **Generative AI-powered APIs**, improving application performance and long-term maintainability.

Associate Analyst, Extremum Analytics

Jul 2024 – Feb 2025 | Gurgaon, India

- Built and deployed **RESTful APIs with FastAPI and Flask**, enabling production-grade integration of Generative AI functionality into live business systems.
- Designed **Azure-based cloud solutions** for secure, scalable data handling across production workloads.
- Automated manual data processing pipelines, improving operational efficiency by **30%+** and freeing analyst time for higher-value work.
- Partnered with stakeholders to debug production issues, improving system reliability and reducing recurring incidents.

PROJECTS

Letter Lyzer - Configuration-Driven Data Extraction Pipeline - Python, JSON Config, Text Parsing

- Engineered a configuration-driven extraction pipeline for a banking client, **cutting manual processing effort by 80%** across 20+ applications using dynamic JSON-based logic.
- Automated cross-referencing of extracted account IDs against CSV datasets, eliminating manual reconciliation and ensuring accurate data mapping.
- Centralized business logic into a scalable architecture, reducing code dependency and cutting onboarding time for new document formats.

Automated OCR Document Processing System - Python, FastAPI, Flask, Azure OCR, Blob Storage

- Built automated document pipelines using **Azure OCR and Generative AI services**, improving document processing efficiency by **75%**.
- Built REST APIs for document validation and frontend integration, enabling real-time processing status across the workflow.

EDUCATION

B.Tech. in Computer Science & Engineering (AI & DS)

2020 – 2024

Panipat Institute of Engineering and Technology | CGPA: 9.04

CERTIFICATIONS & ACHIEVEMENTS

- LangChain: Develop LLM-Powered Applications - Udemy / TCS (2025) | Intro to AI Applications: Vertex AI & Agent space - Udemy / TCS (2026)
- Digital Python E2 - TCS Learning (2026) | Python for Data Science - IBM Developer Skills Network (2022)
- AI Idea Igniter, TCS Internal Innovation Challenge (2025) | Batch Topper & Student of the Batch - CSE (AI & DS), PIET (2024)